5.2	**The sources, origins, physical and working properties of components and systems and their social and ecological footprint**	To apply knowledge and understanding of the advantages, disadvantages and applications of the following components, in order to be able to discriminate between them and select appropriately.
		5.2.1 Sensors: a light-dependent resistors (LDRs) (in topic 1) b thermistor (in topic 1) c moisture sensor d piezoelectric sensor.
		5.2.2 Control devices and components: a rocker switch (on/off) (in topic 1) b resistors (in topic 1) c push to make switch (PTM) d micro switch e reed switch f variable resistors g transistor (bipolar) h microprocessor i microcontroller/PIC j relay.
		5.2.3 Outputs: a buzzers (in topic 1) b light-emitting diodes (LEDs) (in topic 1) c loudspeakers d motors.

SYSTEMS

Key idea	What students need to learn:
	5.2.4 Sources and origins – where components and systems are resourced/manufactured and their geographical origin: a Russia, Saudi Arabia, United States: polymers from crude oil – acrylic, high impact polystyrene (HIPS), acrylonitrile butadiene styrene (ABS) b China, Russia, USA – silicon c China, Australia, Russia – gold d Chile, China, Peru – copper e Australia, Chile, Argentina – lithium f China, Russia, Canada – aluminium g China, Australia, USA – Rare Earth Elements (REEs) h Philippines, Indonesia, Russia, Canada and Australia – nickel.
	5.2.5 The physical characteristics of each component and system: a tolerances, ratings and values – resistor colour codes b material selection for case construction – physical/working properties, sustainability, manufacturing processes.
	5.2.6 Working properties – the way in which each material behaves or responds to external sources: a conductors, insulators – thermal, electrical b polymers used for cases – durability, hardness, toughness, elasticity.
	5.2.7 Social footprint: a relying on scarce and/or hazardous elements used in components and systems – cobalt, tantalum, lithium b effects of using components and systems, including modern communications – mobile phones, computers, games consoles, social media networks.
	5.2.8 Ecological footprint: a effects of material extraction and processing of elements b effects of built-in obsolescence c effects of use d the effects of disposal of components and systems – toxicity of metals and polymers.

SYSTEMS

Key idea	What students need to learn: